

Original Report

Prospective contouring rounds: A novel, high-impact tool for optimizing quality assurance



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Abstract

Purpose: This study was designed to present the results of a novel prospective contouring rounds (CR), in which peer review occurs once the contours and written directive are completed but before initiation of treatment planning.

Methods and materials: Beginning in 2012, all patients undergoing conventionally fractionated radiation therapy at a high-volume academic center were reviewed in a newly initiated daily, prospective, multidisciplinary CR. Cases were scheduled for presentation 2 days after simulation with the expectation that contours would be complete. The clinical suitability of the clinical plan, prescription, contours, and written directive were evaluated and recorded in a prospective database. Treatment planning did not commence until CR approval. Patient information and the prospective database from the first 6 months since program inception, which represented 581 consecutive treatment plans, were pooled and analyzed retrospectively to determine the impact of the prospective peer review at this stage of care delivery.

Results: Sixty-four percent of cases were completed on time without correction. The remaining 36% of cases required modification before treatment planning was initiated. Incomplete contours, target-volume modifications, and alterations to the written directive were the most common corrections or reasons for delay. Decreasing rates of incomplete contours, contour modifications, and miscellaneous delays were seen over time as the program became established. The percentage of cases that had no delays or modifications increased continuously as the program matured in the first 6 months, from 59% to 70%.

Conclusions: Prospective CR is a meaningful and impactful tool in the quality assurance process. More than one-third of cases required contour, directive, or scheduling modification. The establishment of CR improved quality of care, with the percentage of timely, errorless cases increasing steadily over time. The impact of clinical peer review may be optimized by implementation at this early stage of delivery of care rather than at the time of traditional chart rounds.

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Introduction

There is an emerging emphasis on quality assurance (QA) in radiation oncology to ensure the safe and effective delivery of therapeutic radiation and to properly measure the effectiveness of any intervention.¹ An essential

component of the QA process is peer review, whereby one's work is reviewed by professionals of similar standing working in the same field. The majority of radiation oncologists believe that peer review is beneficial to their practice; however, extra resources are necessary to effectively implement a meaningful QA process, and more than 50% of radiation oncologists find it moderately or extremely difficult to incorporate peer review into their schedules.² A recent survey revealed that 83% of radiation oncologists are involved in a peer review process typically referred to as *chart rounds*, an amalgam of chart review, decision making, and detailed treatment plan review. Only 75% of radiation oncologists are comfortable with their peer review program.³ There have been multiple initial reports of successful contouring rounds (CR) programs that have led to improvements in the quality of care.⁴⁻⁶

Current QA processes that are widely adopted typically take place after treatment planning is complete. At best, the data from the Hoopes et al³ suggest that one-third of patients undergo prospective peer review and only 65% have peer review by the end of the first week of treatment, yet 90% of radiation oncologists report that they have changed treatment plans because of the peer review process. Such changes in the contours or written directive require replanning and modification of an active treatment plan, changes to which represent not only suboptimal patient care but also an unnecessary and wasteful deployment of physician and treatment planning time and effort. This is neither efficient nor timely, and it puts a significant strain on the process of care when complex planning is rushed, placing the patient at risk.

Within this context, we present the results of a novel prospective CR in which peer review for each external beam radiation case occurs once the contours and written directive are completed but before the initiation of treatment planning. It was hypothesized that peer review at this point of care delivery may be more impactful than that at the time of traditional chart rounds.

Methods and materials

Starting in June 2012, daily prospective CR were initiated at a high-volume, university-based academic medical center. Unlike traditional chart rounds, cases were presented once contours and fields were completed by the treating physician but before initiation of the planning process.

Our institutional treatment planning process was conducted as described and published elsewhere.⁷ Briefly, a prescription was entered into the MOSAIQ patient information management system (Elekta Corporation, Stockholm, Sweden) and was associated with a corresponding written directive before simulation was scheduled. The written directive served as a clinical care pathway document meant to homogenize institutional practice to follow national best practices per the

evidence-based literature. Care pathways were developed using the Institute of Medicine outline for guideline development.⁸ The written directive covered the breadth of the patient care experience, including guidelines for the prescription, simulation, and treatment planning; on-treatment delivery guidelines, including filming rules; on-treatment clinical management; and follow-up/survivorship. Contouring atlases from national bodies such as the Radiation Therapy Oncology Group were integrated into the directives when available. After simulation, relevant image sets for treatment planning were contoured and reviewed in 1 of several treatment planning systems based on clinical need (Velocity [Varian Medical Systems, Palo Alto, California], Pinnacle [Phillips, Amsterdam, Netherlands], Eclipse [Varian Medical Systems], or iPlan [Brainlab, Feldkirchen, Germany]).

CR were attended by resident physicians, attending physicians, medical students, medical physicists, dosimetrists, nurses, and radiation therapists. All physicians were expected to attend daily unless precluded by tumor board or procedural commitments. Cases were presented in CR 2 days after simulation. The clinical suitability for radiation therapy, the contours (or treatment fields for 2-dimensional cases), and the written directive were reviewed for appropriateness. Cases were re-presented at CR if contours were not complete or a target-volume or field change was recommended. Treatment planning for any case did not commence until CR approval was obtained.

Recommended changes from peer review were recorded on an electronic whiteboard as described previously.⁹ The mechanics of introducing and sustaining CR effectively in our multisite department were simplified substantially by the in-house development and adaptation of this whiteboard. Our whiteboard was developed based on specific design needs by clinicians to serve as an operational platform with electronic links to MOSAIQ and our treatment planning systems. It also serves as a repository for logging incidents observed during CR and as a platform for automating downstream planning assignments and managing the data elements required for subsequent quantitative performance analyses. All data in this study were automatically generated with the whiteboard, thus preserving data integrity relative to otherwise laborious and error-prone manual approaches.

In this study, prospectively collected data from the first 6 months (581 consecutive external beam radiation cases) were analyzed retrospectively after we obtained institutional review board approval. Only plans for conventionally fractionated external beam radiation therapy were included in this study. Data for brachytherapy and stereotactic body radiation therapy are reported separately. Each case was reviewed and scored into as many of the following 7 categories as appropriate, as summarized in Table 1: (1) On time with no correction, (2) non-target-volume change (changes to organs at risk), (3) target-volume change (changes to any target volume), (4) written directive discrepancy (inconsistencies in the written directive or discrepancies between the prescription and written directive), (5) volumes not complete, (6) treatment

Table 1 Scoring categories for contouring rounds review

Category description	Comments/examples
On time, no correction	Case complete and correct
Non-target-volume change	Changes to organs at risk
Target-volume change	Changes to target volume
Written directive change	Inconsistencies in the prescription or care pathway
Volumes incomplete	Volumes not completed
Procedural error	Fusion not done/scan not loaded/simulation not correct
Miscellaneous delay	Imaging pending/coordination of care

planning procedural error (for example, simulation performed incorrectly, simulation scan not loaded into treatment planning system, or physician-requested fusion not completed), or (7) miscellaneous delay (coordination of care issues). Results were pooled and trends analyzed to determine the impact of daily prospective CR on rates of error, physician behavior, and performance over time.

Results

Among the 581 consecutive cases reviewed over the first 6 months of the CR program, 597 total events were registered, and 15 cases were assigned to 2 or more categories. The median number of cases reviewed per day was 7, with the average time per case slightly greater than 8 minutes. The most categories assigned was 3, which happened in only 1 circumstance. Overall events by category are listed in

Table 2 and represented in Figure 1. Sixty-four percent of cases reviewed were on time with no correction necessary. For the 36% of cases that failed CR review and were not immediately ready for treatment planning, the most common reasons were incomplete volumes (11%), inconsistencies in prescription or written directive (9%), or changes required in the target volume (9%). The remaining categories of non-target-volume modification, procedural error, and miscellaneous delay all had error rates that were $\leq 3\%$.

Figure 2 lists the trends in category scoring as a function of time. When scores were analyzed chronologically since CR inception in successive 2-month intervals, the incidence of cases that were on time with no corrections needed increased over the first 6 months from program inception, from 59% to 65% to 70%. Within each subcategory, there was no clear, consistent pattern of change over time, but in general, decreasing rates of contour modifications, incomplete contours, and miscellaneous delays were seen over time as the program became established.

Discussion

We demonstrate here that prospective CR appears to be a meaningful and impactful tool in the QA process. In the first 6 months after inception of a prospective, peer-reviewed CR, more than one-third of cases required a contour, directive, or scheduling modification before treatment planning could proceed. CR improved quality of care, with the percentage of timely, errorless cases steadily improving with time. These results suggest that the impact of clinical peer review may be optimized by

Table 2 Prospective peer review scores

Categories	First 2 mo	Middle 2 mo	Last 2 mo	Grand total	SD, %
One category only					
On time, no correction	116 (59)	168 (65)	99 (70)	383 (64)	5
Non-target-volume change	6 (4)	4 (3)	4 (4)	14 (3)	4
Target-volume change	21 (13)	12 (6)	11 (10)	44 (9)	0
Written directive change	14 (10)	17 (9)	12 (10)	43 (9)	4
Volumes incomplete	18 (10)	34 (14)	9 (6)	61 (11)	0
Procedural error	4 (2)	8 (3)	1 (1)	13 (2)	1
Miscellaneous delay	6 (3)	2 (1)	0 (0)	8 (1)	2
Multiple categories					
Non-target-volume change and target-volume change	1 (1)	0 (0)	1 (1)	2 (0)	0
Non-target-volume change and target-volume change and written directive change	0 (0)	1 (0)	0 (0)	1 (0)	0
Non-target-volume change and written directive change	0 (0)	2 (1)	0 (0)	2 (0)	0
Target-volume change and written directive change	4 (2)	2 (1)	2 (1)	8 (1)	1
Written directive change and volumes incomplete	1 (1)	1 (0)	0 (0)	2 (0)	0
Total	191	251	139	581	

Values are n (%) or %.
SD, standard deviation.

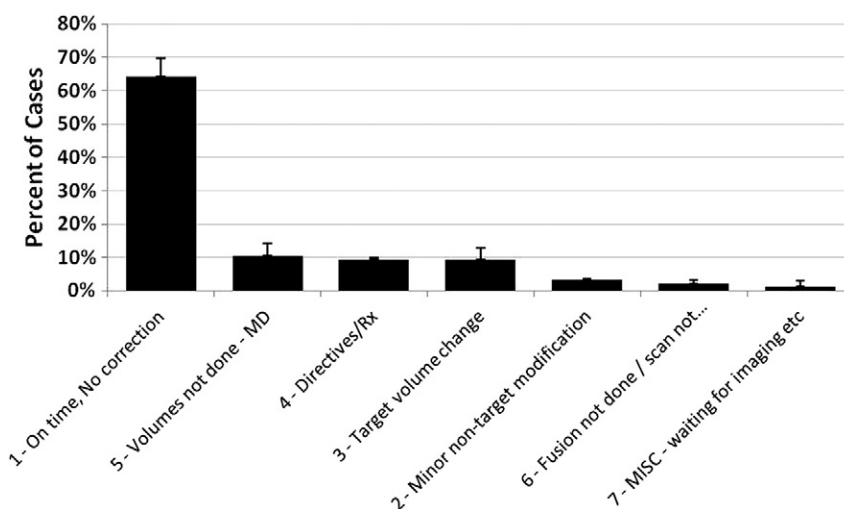


Figure 1 The distribution of contouring rounds scoring during the first 6 months of implementation. Sixty-four percent of cases were complete and correct. The top 3 reasons for cases being incomplete or incorrect were incomplete volumes (11%), inconsistencies in prescription or written directive (8%), and changes required in the target volume (8%). MD, medical doctor; MISC, miscellaneous; Rx, prescription.

implementation at this early stage of delivery of care rather than at the time of traditional chart rounds.

The need for enhanced peer review to optimize quality and safety in radiation oncology is well known. The American Society for Radiation Oncology (ASTRO) has called for identification and development of efficient peer review techniques to optimize the safety and quality of radiation therapy treatments.¹⁰ Although cooperative research groups have integrated QA procedures into protocols for more than 40 years,¹¹ QA procedures are unevenly used in modern practice in terms of implementation and use.^{2,10} The consequences of not adhering to proper radiation treatment planning procedures have been well reported. Several analyses of cooperative group trials have demonstrated the detrimental effects of noncompliance with proper contouring and treatment planning

procedures.^{12,13} Ohri et al¹ reported a meta-analysis of cooperative group trials and demonstrated that radiation therapy protocol deviations are associated with increased risks of treatment failure and overall mortality. Between 8% and 71% of national protocol cases had deviations that led to a 75% increase in the risk of treatment failures and overall mortality.¹ Importantly, the process of centralized review before treatment initiation has been shown to minimize treatment deviations for national protocols.^{14,15}

Retrospective and prospective studies deploying systems engineering approaches in radiation medicine have indicated a strong coupling between select stages of the treatment planning process and patient safety outcomes.⁹ Specifically, the contouring and prescription steps were shown to be among the highest patient-safety risk steps in the planning process. In response, we developed CR to

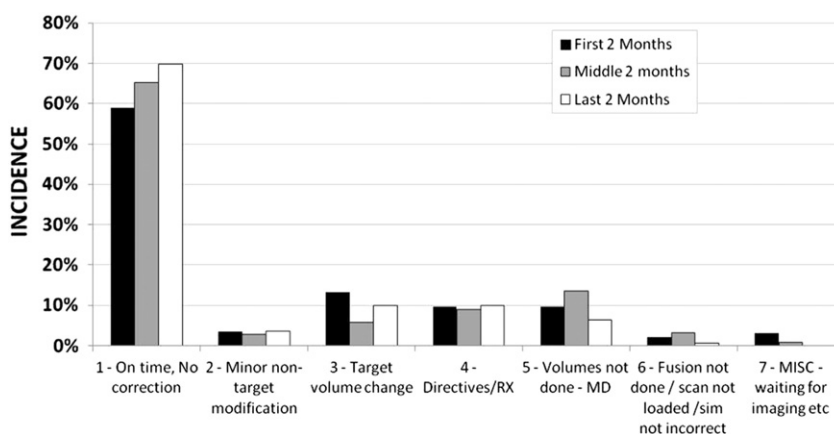


Figure 2 Trends in category scoring as a function of time. Percentage incidences for each category are shown. Note the incidence of cases that were on time with no corrections needed increased continually over the first 6 months from program inception. MD, medical doctor; MISC, miscellaneous; RX, prescription; sim, simulation.

ensure greater scrutiny on these steps by developing a library of evidence-based directives that encompass the treatment planning spectrum⁴ and enforcing multidisciplinary peer review at a more optimal inflection time point within the process, namely before initiation of treatment planning.

Our implementation of daily CR was thus meant to address the challenges inherent in maintaining a high-quality, efficient QA process in a high-volume academic practice with increasingly complex radiation therapy techniques becoming standard, including intensity modulated radiation therapy, image guided radiation therapy, and stereotactic body radiation therapy. Target volumes are reviewed carefully, and dose-fractionation schemes are considered for the reasons highlighted above. Additionally, the overall appropriateness of radiation therapy is an important consideration, particularly when the concept of value-based care and cost efficiency will be increasingly emphasized in the delivery of modern radiation therapy services.¹⁶ Review at this point does allow for cancellation of a planned course of radiation before it is administered.

We hypothesized that CR would increase the quality of care in an efficient manner by creating a culture of safety while still prioritizing physician time, a process called “smarter radiation oncology.” Inherent in this process was the clear expectation that physicians were obligated to participate in CR. Active and regular commitment of time and participation is mandated from all faculty and resident physicians, and clinical activities are structured to permit and protect this time. Our department consists of 14 physicians and 8 resident physicians spread over 8 different physical sites spanning a large regional area, ranging from high-volume tertiary care medical centers to smaller regional hospitals. We surmounted this physical challenge by having daily teleconferences with all sites, with full 2-way audio and visual communication. The culture of CR has been carefully groomed to promote open professional communication and acceptance of constructive criticism. Our department embraces the concept that some small percentage of modifications are likely inevitable given the complex scope of modern practice. Physicians are expected to vigilantly work to minimize the amount of modifications, but it is emphasized and understood that modifications represent important improvements in patient care that optimize the quality and safety of their individual practices.

The performance of daily multidisciplinary peer review before treatment planning commences is an optimal way to ensure compliance with standard, evidence-based department directives. It also circumvents some of the potential limitations of traditional weekly chart rounds, including not being able to take the time to review the contouring on a methodical, slice-by-slice basis because of the overall volume of cases. These time constraints run the risk of leading to a cursory and superficial review that leaves little opportunity to catch errors or recommend changes to a complex case and treatment plan.

CR also offers several potential advantages in terms of operational efficiency. If changes to the treatment plan are recommended at the time of traditional chart rounds, there are 2 significant consequences. First, it is inherent that suboptimal care was delivered for a portion of the treatment, with the increased potential for reduced outcomes. Second, any changes in a treatment plan require a duplication of work by the physician, planners, and therapists involved in the case as the remediated plan is processed and deployed. With remediated plans, there is the increased risk of delivering delay-rushed care when the perceived need to expedite treatment initiation dominates over QA considerations. Therefore, the process of CR may be more cost and time efficient, can prevent incorrect fractions from even being given, and may reduce the medical errors associated with replanning.

Our findings that a substantial portion of cases needed modification before treatment planning are in line with the findings of other institutions. MD Anderson reported the results of a prospective peer review conference in which the plan was reviewed after completion for nonpalliative cases.¹⁷ They found that 12% of cases required a change (dose change in 28.3% of modified cases, target-volume change in 69.1% of cases, and a major treatment change in 2.6% of cases). They also found that the rates of necessary changes decreased over a 4-year time period and recommended that peer review should be a national requirement. Our study is different in that other considerations besides the contours and treatment plan were included and categorized (eg, physician delinquency with contours, fusion errors, and directive compliance), and all conventionally fractionated radiation cases were included. Our study also suggests improvements in care from a prospective CR program can be noted over a much shorter period of time (6 months vs 4 years).

The limitations of CR include the resource-intensive requirements of the multidisciplinary team to run daily CR instead of weekly chart rounds. This was required to provide the optimal time to review each case in detail. As such, we demonstrated significantly higher rates of case modification than the 4% to 7% reported by Hoopes et al³ (Table 2). Furthermore, we recognized that waiting for CR approval for treatment planning would potentially delay the start of a patient's treatment. This tradeoff was accepted as a component of the department's “no-fly policy” that creates interlocks for the process steps that lead up to the start of treatment.¹⁵

Future directions include the implementation of plan review to the CR to provide peer review of the treatment plan. We also plan to determine the impact of CR on operational efficiency, using CR to identify metrics for practice quality improvements required for physician maintenance of certification and examining rates of modifications as a function of physician, treatment modality, and treatment site. We have already started to expand the scope of chart rounds to include all aspects of

our treatment modalities, including initiating programs for stereotactic radiosurgery, stereotactic body radiation therapy, and brachytherapy. We intend to eventually develop both a physician evaluation system based on chart round performance and a resident evaluation system to track the evolution of resident technical aptitude over time and document achievement of Accreditation Council for Graduate Medical Education-mandated core competencies. Finally, we are working to ensure that the CR system is easily scalable to meet the future needs of the department and health system.

In summary, CR is a comprehensive, multidisciplinary peer-to-peer review that occurs at the beginning of the treatment planning process. CR successfully identified errors or required changes in one-third of cases treated at a high-volume academic institution. Implementation of the CR program led to improvements in the quality of care, with the percentage of timely, errorless radiation treatment plans steadily increasing over the first 6 months after implementation. The CR program allows for a more meaningful review of the patient's treatment plan via a formalized discussion of care between physicians, medical physicists, dosimetrists, and radiation therapists at the beginning of the radiation planning process.

References

1. Ohri N, Shen X, Dicker AP, Doyle LA, Harrison AS, Showalter TN. Radiotherapy protocol deviations and clinical outcomes: A meta-analysis of cooperative group clinical trials. *J Natl Cancer Inst*. 2013;105:387-393.
2. Hamilton SN, Hasan H, Parsons C, et al. Canadian radiation oncologists' opinions regarding peer review: A national survey. *Pract Radiat Oncol*. 2015;5:120-126.
3. Hoopes DJ, Johnstone PA, Chapin PS, et al. Practice patterns for peer review in radiation oncology. *Pract Radiat Oncol*. 2015;5:32-38.
4. Joye I, Lambrecht M, Jegou D, Hortobágyi E, Scalliet P, Haustermans K. Does a central review platform improve the quality of radiotherapy for rectal cancer? Results of a national quality assurance project. *Radiother Oncol*. 2014;111:400-405.
5. Lymberiou T, Galuszka S, Lee G, et al. Predictors of breast radiotherapy plan modifications: Quality assurance rounds in a large cancer centre. *Radiother Oncol*. 2015;114:17-21.
6. Chera BS, Mazur L, Jackson M, et al. Quantification of the impact of multifaceted initiatives intended to improve operational efficiency and the safety culture: A case study from an academic medical center radiation oncology department. *Pract Radiat Oncol*. 2014;4:e101-e108.
7. Potters L, Raince J, Chou H, et al. Development, implementation, and compliance of treatment pathways in radiation medicine. *Front Oncol*. 2013;3:105.
8. Kapur A, Potters L. Six sigma tools for a patient safety-oriented, quality-checklist driven radiation medicine department. *Pract Radiat Oncol*. 2012;2:86-96.
9. Kapur A, Goode G, Riehl C, et al. Incident learning and failure-mode-and-effects-analysis guided safety initiatives in radiation medicine. *Front Oncol*. 2013;3:305.
10. Marks LB, Adams RD, Pawlicki T, et al. Enhancing the role of case-oriented peer review to improve quality and safety in radiation oncology: Executive summary. *Pract Radiat Oncol*. 2013;3:149-156.
11. Pawlicki T, Chera B, Ning T, Marks LB. The systematic application of quality measures and process control in clinical radiation oncology. *Semin Radiat Oncol*. 2012;22:70-76.
12. Abrams RA, Winter KA, Regine WF, et al. Failure to adhere to protocol specified radiation therapy guidelines was associated with decreased survival in RTOG 9704: A phase III trial of adjuvant chemotherapy and chemoradiotherapy for patients with resected adenocarcinoma of the pancreas. *Int J Radiat Oncol Biol Phys*. 2012;82:809-816.
13. Peters LJ, O'Sullivan B, Giralt J, et al. Critical impact of radiotherapy protocol compliance and quality in the treatment of advanced head and neck cancer: Results from TROG 02.02. *J Clin Oncol*. 2010;28:2996-3001.
14. Bentzen SM, Bernier J, Davis JB, et al. Clinical impact of dosimetry quality assurance programmes assessed by radiobiological modelling of data from the thermoluminescent dosimetry study of the European Organization for Research and Treatment of Cancer. *Eur J Cancer*. 2000;36:615-620.
15. Ibbott GS, Hanson WF, O'Meara E, et al. Dose specification and quality assurance of radiation therapy oncology group protocol 95-17: A cooperative group study of iridium-192 breast implants as sole therapy. *Int J Radiat Oncol Biol Phys*. 2007;69:1572-1578.
16. Hayes JH, Ollendorf DA, Pearson SD, et al. Observation versus initial treatment for men with localized, low-risk prostate cancer: A cost-effectiveness analysis. *Ann Intern Med*. 2013;158:853-860.
17. Ballo MT, Chronowski GM, Schlembach PJ, Bloom ES, Arzu IY, Kuban DA. Prospective peer review quality assurance for outpatient radiation therapy. *Pract Radiat Oncol*. 2014;4:279-284.